FURTHER MECHANICS / KINEMATICS & DYNAMICS

Motion Under a Variable Force

Worked solutions 14 questions · 162 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*9 marks*	2
Question 2	*7 marks*	4
Question 3	*9 marks*	5
Question 4	*13 marks*	7
Question 5	*14 marks*	9
Question 6	*16 marks*	11
Question 7	*9 marks*	13
Question 8	*11 marks*	15
Question 9	*12 marks*	17
Question 10	*10 marks*	19
Question 11	*13 marks*	21
Question 12	*9 marks*	23
Question 13	*14 marks*	25
Question 14	*16 marks*	27

1. A particle P of mass $4\,\text{kg}$ moves along a straight horizontal line. At time $t\,\text{s}$, its displacement in the positive direction from a fixed point O is $x\,\text{m}$ and its velocity is $v\,\text{m}\,\text{s}^{-1}$.

The only horizontal force acting on P has magnitude

$$\frac{36}{(x+2)^3}\,\text{N}$$

and acts in the direction of decreasing x. Initially, P is at O and is moving in the positive direction with speed $\frac{3}{2}\,\text{m}\,\text{s}^{-1}$.

(a) Find an expression for v in terms of x. **[5]**

(b) Hence find an expression for x in terms of t. **[4]**

Solution

(a) Take the direction of increasing x as positive. Newton's second law gives

$$4a = -\frac{36}{(x+2)^3}$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, then separating variables and integrating:

$$4v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{36}{(x+2)^3}$$

$$\int 4v\,\mathrm{d}v = -36\int (x+2)^{-3}\,\mathrm{d}x$$

$$2v^2 = \frac{18}{(x+2)^2} + c$$

Initially, $x = 0$ and $v = \frac{3}{2}$, so

$$2\left(\frac{3}{2}\right)^2 = \frac{18}{2^2} + c \quad \Longrightarrow \quad c = 0$$

Thus $v^2 = \frac{9}{(x+2)^2}$. The velocity is initially positive and cannot become zero at any finite $x \geq 0$, so we take the positive root:

$$v = \frac{3}{x+2}$$

(b) Using part (a) and $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, separate variables and integrate:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3}{x+2}$$

$$\int (x+2)\,\mathrm{d}x = \int 3\,\mathrm{d}t$$

$$\frac{x^2}{2} + 2x = 3t + d$$

Since $x = 0$ when $t = 0$, $d = 0$. Completing the square gives

$$x^2 + 4x = 6t$$

$$(x + 2)^2 = 6t + 4$$

Taking the positive root since $x + 2 > 0$:

$$x = \sqrt{6t + 4} - 2, \qquad t \geq 0$$

2. A lorry of mass 2000 kg travels along a straight horizontal road. Its engine delivers constant power of 80 kW. When the lorry is moving at speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude $200v\,\mathrm{N}$.

At time t seconds, the speed of the lorry is $v\,\mathrm{m\,s^{-1}}$. At $t = 0$, the speed of the lorry is $8\,\mathrm{m\,s^{-1}}$.

(a) Show that

$$10v\frac{\mathrm{d}v}{\mathrm{d}t} = 400 - v^2$$ **[3]**

(b) Verify that, while $8 \leq v < 20$,

$$t = 5\ln\left(\frac{336}{400 - v^2}\right)$$ **[4]**

Solution

(a) Take the direction of motion as positive. The driving force is

$$\frac{P}{v} = \frac{80\,000}{v}\,\mathrm{N}$$

Using acceleration $\frac{\mathrm{d}v}{\mathrm{d}t}$, Newton's second law gives

$$2000\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{80\,000}{v} - 200v$$

Multiplying by $\frac{v}{200}$:

$$10v\frac{\mathrm{d}v}{\mathrm{d}t} = 400 - v^2$$

(b) Separating the variables in the equation from part (a):

$$\frac{10v}{400 - v^2}\,\mathrm{d}v = \mathrm{d}t$$

Since the derivative of $400 - v^2$ is $-2v$, integrating gives

$$\int 1\,\mathrm{d}t = -5\int \frac{-2v}{400 - v^2}\,\mathrm{d}v$$

$$t = -5\ln|400 - v^2| + c$$

For $8 \leq v < 20$, $400 - v^2 > 0$, so the modulus can be dropped. Using $v = 8$ when $t = 0$:

$$0 = -5\ln(400 - 8^2) + c$$

$$c = 5\ln 336$$

Substituting and combining the logarithms:

$$t = 5\ln 336 - 5\ln(400 - v^2)$$

$$= 5\ln\left(\frac{336}{400 - v^2}\right)$$

3. A particle P of mass $0.5\,\text{kg}$ moves along the positive x-axis under the action of a single force. At time t seconds, the particle is x metres from the origin O, where $x > 0$, and has speed $v\,\text{m s}^{-1}$. The force acts in the direction of increasing x and has magnitude

$$\frac{k}{x^3} \text{ newtons}$$

where k is a positive constant.

When $x = 2$, the speed of P is $1\,\text{m s}^{-1}$, and when $x = 4$, its speed is $2\,\text{m s}^{-1}$.

(a) Show that

$$v^2 = 5 - \frac{16}{x^2} \qquad \textbf{[5]}$$

The time taken for the speed of P to increase from $1\,\text{m s}^{-1}$ to $2\,\text{m s}^{-1}$ is T seconds.

(b) Use algebraic integration to show that

$$T = \frac{6}{5} \qquad \textbf{[4]}$$

Solution

(a) Take the direction of increasing x as positive. Newton's second law gives

$$\frac{1}{2}a = +\frac{k}{x^3}$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, and separating the variables:

$$\frac{1}{2}v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{k}{x^3}$$

$$2v\,\mathrm{d}v = 4kx^{-3}\,\mathrm{d}x$$

$$\int 2v\,\mathrm{d}v = \int 4kx^{-3}\,\mathrm{d}x$$

$$v^2 = -\frac{2k}{x^2} + c$$

Substituting the two given conditions:

$$1 = -\frac{k}{2} + c$$

$$4 = -\frac{k}{8} + c$$

Subtracting the first equation from the second:

$$3 = \frac{3k}{8}$$

$$k = 8$$

$$c = 1 + \frac{8}{2} = 5$$

Hence

$$v^2 = 5 - \frac{16}{x^2}$$

(b) Since the particle moves in the direction of increasing x, use the positive square root from part (a). As $x > 0$,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = \sqrt{5 - \frac{16}{x^2}} = \frac{\sqrt{5x^2 - 16}}{x}$$

Separating the variables and integrating:

$$\mathrm{d}t = \frac{x}{\sqrt{5x^2 - 16}}\,\mathrm{d}x$$

$$\int \mathrm{d}t = \frac{1}{10}\int 10x(5x^2 - 16)^{-1/2}\,\mathrm{d}x$$

$$t = \frac{1}{5}\sqrt{5x^2 - 16} + d$$

Let t_1 and t_2 be the times when $x = 2$ and $x = 4$, respectively. Substituting these values:

$$t_1 = \frac{1}{5}\sqrt{20 - 16} + d = \frac{2}{5} + d$$

$$t_2 = \frac{1}{5}\sqrt{80 - 16} + d = \frac{8}{5} + d$$

$$T = t_2 - t_1 = \frac{8}{5} - \frac{2}{5} = \frac{6}{5}\,\mathrm{s}$$

4. A particle P of mass $3\,\text{kg}$ moves along the positive x-axis. At time $t = 0$, it passes through the origin with speed $4\,\text{m}\,\text{s}^{-1}$.

While P is moving, the only force acting on it is a resistive force of magnitude

$$\frac{6}{v}\,\text{N}$$

where $v\,\text{m}\,\text{s}^{-1}$ is its speed. The force acts in the negative x-direction.

(a) Find an expression for v in terms of the displacement x of P from the origin. **[5]**

(b) Find the distance travelled by P while its speed decreases from $4\,\text{m}\,\text{s}^{-1}$ to $2\,\text{m}\,\text{s}^{-1}$. **[2]**

A particle Q, also of mass $3\,\text{kg}$, starts from the origin with speed $4\,\text{m}\,\text{s}^{-1}$. The only force acting on Q is a constant resistive force of magnitude $\frac{3}{2}\,\text{N}$.

(c) By comparing the resistive forces acting on P and Q, explain why P must come to rest at a time $t < 8\,\text{s}$ and at a displacement $x < 16\,\text{m}$. **[3]**

(d) Sketch the velocity–time graph for P from $t = 0$ until it first comes to rest. Your sketch should indicate the concavity of the graph, but no numerical values are required. **[1]**

(e) Determine the maximum displacement of P from the origin. **[2]**

Solution

(a) Take the positive direction to be along the positive x-axis. Newton's second law gives

$$3a = -\frac{6}{v}$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$3v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{6}{v}$$

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = -2$$

Separating the variables and integrating:

$$\int v^2\,\mathrm{d}v = -2\int \mathrm{d}x$$

$$\frac{v^3}{3} = -2x + c$$

When $x = 0$, $v = 4$, so $c = \frac{64}{3}$. Hence

$$v^3 = 64 - 6x$$

$$v = \sqrt[3]{64 - 6x}$$

(b) Substituting $v = 2$ into the result of part (a):

$$8 = 64 - 6x \quad \Longrightarrow \quad x = \frac{28}{3}\,\mathrm{m}$$

Since P moves in the positive direction throughout, the distance travelled is $\frac{28}{3}\,\mathrm{m}$.

(c) For Q, Newton's second law in the same positive direction gives

$$3a_Q = -\frac{3}{2} \quad \Longrightarrow \quad a_Q = -\frac{1}{2}\,\mathrm{m\,s^{-2}}$$

Using constant-acceleration formulae, its stopping time and displacement are

$$0 = 4 - \frac{1}{2}t \quad \Longrightarrow \quad t = 8\,\mathrm{s}$$

$$x_Q = 4(8) + \frac{1}{2}\left(-\frac{1}{2}\right)(8)^2 = 16\,\mathrm{m}$$

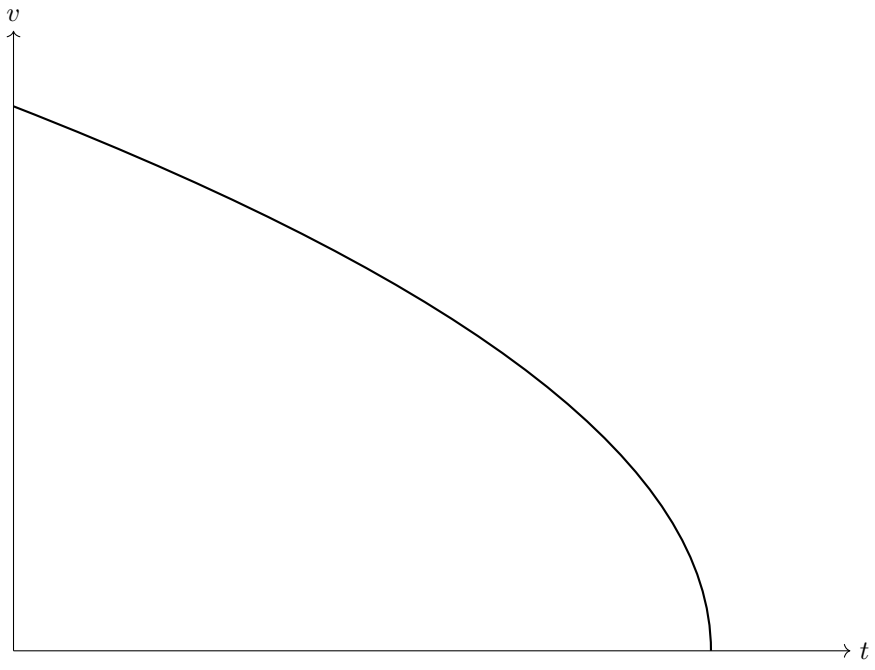
The initial resistive forces are equal. As P slows down, however,

$$0 < v < 4 \quad \Longrightarrow \quad \frac{6}{v} > \frac{3}{2}$$

Thus, at every speed below $4\,\mathrm{m\,s^{-1}}$, P has a greater deceleration than Q. Each further decrease in speed therefore takes less time and covers less distance for P, so it comes to rest with

$$t < 8\,\mathrm{s}, \qquad x < 16\,\mathrm{m}$$

(d) The sketch should look like:



As the speed falls, the resistive force increases, making the acceleration more negative. The graph is therefore decreasing and concave down.

(e) The displacement increases while P is moving, so its maximum occurs as P comes to rest. Taking $v \to 0$ in part (a):

$$0 = 64 - 6x_{\max} \quad \Longrightarrow \quad x_{\max} = \frac{32}{3}\,\mathrm{m}$$

5. A particle B of mass 2 kg moves along the positive x-axis. When B passes through the origin, its speed is $u\,\mathrm{m\,s^{-1}}$.

The only horizontal force acting on B is a resistance whose magnitude is directly proportional to the cube of the speed of B. When the speed of B is $3\,\mathrm{m\,s^{-1}}$, the magnitude of the resistance is 54 N.

(a) Show that, while B is moving in the positive x-direction, its motion satisfies

$$\frac{1}{v^2}\frac{\mathrm{d}v}{\mathrm{d}x} = -1$$ **[3]**

(b) (i) Solve the differential equation in part (a) to obtain v in terms of x and u. **[4]**

(ii) By considering the behaviour of v as $x \to \infty$, state one respect in which this model of the motion is unrealistic. **[1]**

The particle subsequently reaches a point A with coordinate X, where its speed is $V\,\mathrm{m\,s^{-1}}$. The work done by the resistance as B moves from the origin to A is W J.

(c) (i) Using $W = \int F\,\mathrm{d}x$, where F is the signed force in the x-direction, find an expression for W in terms of X and u. **[3]**

(ii) Explain the physical significance of the sign of W. **[1]**

(iii) Rewrite your answer to part (c)(i) in terms of V and u, and hence show how W is related to the change in the kinetic energy of B. **[2]**

Solution

(a) Take the positive direction to be the positive x-direction. Let the magnitude of the resistance be kv^3. The given force and speed give

$$54 = k(3^3) \quad \Rightarrow \quad k = 2$$

Newton's second law in the positive x-direction is

$$2a = -kv^3 = -2v^3$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -2v^3$$

While the particle is moving in the positive direction, $v > 0$, so dividing by $2v^3$ gives

$$\frac{1}{v^2}\frac{\mathrm{d}v}{\mathrm{d}x} = -1$$

(b) (i) Separating the variables and integrating:

$$\int v^{-2}\,\mathrm{d}v = -\int 1\,\mathrm{d}x$$

$$-\frac{1}{v} = -x + c$$

At the origin, $x = 0$ and $v = u$, so $c = -\frac{1}{u}$. Hence

$$-\frac{1}{v} = -x - \frac{1}{u}$$
$$\frac{1}{v} = x + \frac{1}{u} = \frac{1+ux}{u}$$
$$v = \frac{u}{1+ux}$$

(ii) As $x \to \infty$, $v \to 0$, but $v > 0$ for every finite x. Thus the model unrealistically predicts that the particle travels an unlimited distance without coming to rest.

(c) (i) The signed force is $F = -2v^3$. Using the speed from part (b)(i):

$$W = \int_0^X F \,\mathrm{d}x$$
$$= -2u^3 \int_0^X (1+ux)^{-3} \,\mathrm{d}x$$
$$= -2u^3 \left[\frac{(1+ux)^{-2}}{-2u}\right]_0^X$$
$$= u^2\left[(1+uX)^{-2} - 1\right]$$

(ii) $W < 0$: the resistance acts opposite to the displacement, so it does negative work and reduces the particle's kinetic energy.

(iii) From part (b)(i), $V = \frac{u}{1+uX}$, so the expression in part (c)(i) becomes

$$W = \frac{u^2}{(1+uX)^2} - u^2 = V^2 - u^2$$

Since the mass is 2 kg, the change in kinetic energy is

$$\Delta\mathrm{KE} = \frac{1}{2}(2)V^2 - \frac{1}{2}(2)u^2 = V^2 - u^2 = W$$

6. A small sphere P of mass 5 kg is released into a liquid and moves vertically downwards. Take the downward direction as positive. In a model of the motion, the forces acting on P are:

- its weight;
- a constant upthrust of magnitude 14 N;
- a resistance to motion whose magnitude is proportional to the speed of P.

When the speed of P is $4\,\mathrm{m\,s^{-1}}$, the resistance has magnitude 10 N. At time $t = 0$, the sphere is at a point O and is moving downwards with speed $u\,\mathrm{m\,s^{-1}}$, where $u > 0$. At time t seconds, its downward velocity is $v\,\mathrm{m\,s^{-1}}$. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) Show that, according to the model,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 7 - \frac{1}{2}v$$ **[4]**

(b) (i) Solve the differential equation in part (a) to show that

$$v = 14 + (u - 14)e^{-t/2}$$ **[5]**

(ii) Describe briefly how the speed of P varies over time in each of the following cases:

- $0 < u < 14$;
- $u = 14$;
- $u > 14$.

In each case, state the limiting speed of P as $t \to \infty$. **[3]**

(c) Given that $u = 9$, find the exact downward displacement of P from O when $t = 6$. **[4]**

Solution

(a) Write the resistance as kv. Since $4k = 10$, $k = \frac{5}{2}$.

Taking downwards as positive, Newton's second law gives:

$$5\frac{\mathrm{d}v}{\mathrm{d}t} = 5(9.8) - 14 - \frac{5}{2}v$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 7 - \frac{1}{2}v$$

(b) (i) For $u \neq 14$, separate the variables in the equation from part (a):

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{2}(v - 14)$$

$$\int \frac{1}{v - 14}\,\mathrm{d}v = -\frac{1}{2}\int 1\,\mathrm{d}t$$

$$\ln|v - 14| = -\frac{t}{2} + c$$

Using $v = u$ when $t = 0$ gives $c = \ln|u - 14|$. Hence:

$$\ln|v - 14| - \ln|u - 14| = -\frac{t}{2}$$

$$|v - 14| = |u - 14|e^{-t/2}$$

The right-hand side is non-zero for finite t, so $v - 14$ cannot change sign. It therefore has the same sign as its initial value $u - 14$, giving:

$$v - 14 = (u - 14)e^{-t/2}$$

If $u = 14$, the differential equation gives the constant solution $v = 14$. Thus, for every $u > 0$,

$$v = 14 + (u - 14)e^{-t/2}$$

(ii) Since $e^{-t/2}$ decreases from 1 towards 0:

- If $0 < u < 14$, the speed increases from u towards $14\,\mathrm{m\,s^{-1}}$.
- If $u = 14$, the speed remains constant at $14\,\mathrm{m\,s^{-1}}$.
- If $u > 14$, the speed decreases from u towards $14\,\mathrm{m\,s^{-1}}$.

In all three cases, the limiting speed is $14\,\mathrm{m\,s^{-1}}$.

(c) Let x be the downward displacement from O, so $x = 0$ when $t = 0$. Using $u = 9$ in part (b)(i):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = 14 - 5e^{-t/2}$$

Integrating with respect to t:

$$x = 14t + 10e^{-t/2} + A$$

$$0 = 10 + A$$

$$A = -10$$

Therefore, at $t = 6$, the downward displacement is:

$$x = 14(6) + 10e^{-3} - 10 = \left(74 + 10e^{-3}\right)\mathrm{m}$$

7. A vehicle of mass 900 kg moves along a straight horizontal road. Its engine provides a constant driving force of 576 N. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude (v^2+14v) N.

The time taken for the speed of the vehicle to increase from $3\,\mathrm{m\,s^{-1}}$ to $13\,\mathrm{m\,s^{-1}}$ is T seconds.

(a) Show that

$$T = 18\ln\left(\frac{27}{7}\right)$$ **[6]**

When $t = 0$, the speed of the vehicle is $8\,\mathrm{m\,s^{-1}}$.

(b) Show that

$$\frac{v+32}{18-v} = pe^{qt}$$

where the rational numbers p and q are to be found. **[2]**

(c) Hence explain why $v < 18$ for all $t \geq 0$. **[1]**

Solution

(a) Take the direction of motion as positive. Using $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, Newton's second law gives

$$\begin{aligned} 900\frac{\mathrm{d}v}{\mathrm{d}t} &= 576 - (v^2 + 14v) \\ &= (18-v)(v+32) \end{aligned}$$

Separating the variables:

$$\frac{900}{(18-v)(v+32)}\,\mathrm{d}v = \mathrm{d}t$$

For the partial fractions, write

$$\frac{1}{(18-v)(v+32)} = \frac{A}{18-v} + \frac{B}{v+32}$$

Multiplying by $(18-v)(v+32)$ and comparing coefficients gives

$$\begin{aligned} 1 &= A(v+32) + B(18-v) \\ &= (A-B)v + 32A + 18B \end{aligned}$$

$$A - B = 0, \qquad 32A + 18B = 1$$

$$A = B = \frac{1}{50}$$

Hence, integrating both sides:

$$18\int\left(\frac{1}{18-v} + \frac{1}{v+32}\right)\mathrm{d}v = \int 1\,\mathrm{d}t$$

$$18(-\ln|18-v| + \ln|v+32|) = t + c$$

For $3 \le v \le 13$, both $18 - v$ and $v + 32$ are positive, so

$$18 \ln\left(\frac{v+32}{18-v}\right) = t + c$$

Let t_3 and t_{13} be the times at which the speeds are 3 and $13\,\mathrm{m\,s^{-1}}$, respectively. Substituting these speeds:

$$t_3 + c = 18 \ln\left(\frac{35}{15}\right) = 18 \ln\left(\frac{7}{3}\right)$$

$$t_{13} + c = 18 \ln\left(\frac{45}{5}\right) = 18 \ln 9$$

Subtracting eliminates c:

$$T = t_{13} - t_3 = 18\left(\ln 9 - \ln\frac{7}{3}\right)$$

$$= 18 \ln\left(\frac{27}{7}\right)$$

(b) Using the integrated relation from part (a), the condition $v = 8$ when $t = 0$ gives

$$c = 18 \ln\left(\frac{8+32}{18-8}\right) = 18 \ln 4$$

Thus, rearranging and exponentiating:

$$\ln\left(\frac{v+32}{18-v}\right) = \frac{t}{18} + \ln 4$$

$$\frac{v+32}{18-v} = 4e^{t/18}$$

This is the required form pe^{qt}, with

$$p = 4, \qquad q = \frac{1}{18}$$

(c) From part (b), $v + 32 = 4e^{t/18}(18 - v)$. Since $v + 32 = 50 - (18 - v)$,

$$18 - v = \frac{50}{1 + 4e^{t/18}} > 0$$

The denominator is positive for every $t \ge 0$, so $v < 18$ for all $t \ge 0$.

8. A particle P of mass $2\,\text{kg}$ moves along the positive x-axis in the direction of increasing x. At time t seconds, its displacement from the origin is x metres and its speed is $v\,\text{m s}^{-1}$.

While P is moving, the resultant force acting on it is directed towards the origin and has magnitude $kv(v+2)\,\text{N}$, where k is a positive constant and v denotes the numerical value of the speed in metres per second.

When $x = 1$, the speed of P is $7\,\text{m s}^{-1}$, and when $x = 3$, its speed is $1\,\text{m s}^{-1}$.

(a) Find k and show that

$$v = 9 \cdot 3^{(1-x)/2} - 2$$ **[6]**

The time taken for the speed of P to decrease from $7\,\text{m s}^{-1}$ to $1\,\text{m s}^{-1}$ is T seconds.

(b) Show that

$$T = \frac{\ln(7/3)}{\ln 3}$$ **[5]**

Solution

(a) Take the positive direction to be away from the origin, in the direction of increasing x. The resultant force acts in the negative direction, so Newton's second law gives

$$2a = -kv(v+2)$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -kv(v+2)$$

Since $v > 0$ while the particle is moving, divide by v and separate the variables:

$$\frac{1}{v+2}\,\mathrm{d}v = -\frac{k}{2}\,\mathrm{d}x$$

$$\int \frac{1}{v+2}\,\mathrm{d}v = -\frac{k}{2}\int \mathrm{d}x$$

$$\ln|v+2| = -\frac{kx}{2} + c$$

Here $v + 2 > 0$, so the modulus can be dropped. Substituting the two given conditions:

$$\ln 9 = -\frac{k}{2} + c$$

$$\ln 3 = -\frac{3k}{2} + c$$

Subtracting the first equation from the second:

$$\ln 3 - \ln 9 = -k$$

$$k = \ln 3$$

Subtracting the relation at $x = 1$, $v = 7$ from the general relation now gives

$$\ln(v+2) - \ln 9 = -\frac{k}{2}(x-1)$$

$$\ln\left(\frac{v+2}{9}\right) = \frac{1-x}{2}\ln 3$$

$$\frac{v+2}{9} = 3^{(1-x)/2}$$

$$v = 9 \cdot 3^{(1-x)/2} - 2$$

(b) Using $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, Newton's second law in the same positive direction gives

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = -kv(v+2)$$

Separating the variables, and using partial fractions:

$$\frac{2}{v(v+2)}\,\mathrm{d}v = -k\,\mathrm{d}t$$

$$\frac{2}{v(v+2)} = \frac{1}{v} - \frac{1}{v+2}$$

$$\int\left(\frac{1}{v} - \frac{1}{v+2}\right)\mathrm{d}v = -k\int \mathrm{d}t$$

$$\ln|v| - \ln|v+2| = -kt + d$$

Both v and $v+2$ are positive, so this becomes

$$\ln\left(\frac{v}{v+2}\right) = -kt + d$$

Let t_0 be the time when $v = 7$. Then $v = 1$ at time $t_0 + T$, giving

$$\ln\frac{7}{9} = -kt_0 + d$$

$$\ln\frac{1}{3} = -k(t_0+T) + d$$

Subtracting eliminates d and t_0. Using $k = \ln 3$ from part (a):

$$\ln\frac{1}{3} - \ln\frac{7}{9} = -kT$$

$$\ln\frac{3}{7} = -T\ln 3$$

$$T = \frac{\ln(7/3)}{\ln 3}$$

9. A particle P of mass 4 kg is released from rest and moves vertically downwards through a fluid. While P is moving with speed $v\,\mathrm{m\,s^{-1}}$, it experiences a resistance of magnitude $6v\,\mathrm{N}$ acting vertically upwards. After P has descended a distance $x\,\mathrm{m}$, its speed is $v\,\mathrm{m\,s^{-1}}$. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) Show that

$$\left(\frac{10v}{98-15v}\right)\frac{\mathrm{d}v}{\mathrm{d}x} = 1 \qquad \textbf{[2]}$$

(b) Verify that x and v satisfy

$$45x = 196\ln\left(\frac{98}{98-15v}\right) - 30v \qquad \textbf{[5]}$$

(c) Determine the work done against the resistance from the instant P is released until its acceleration has magnitude $2.45\,\mathrm{m\,s^{-2}}$. Give your answer to 3 significant figures. **[5]**

Solution

(a) Take vertically downwards as positive. Newton's second law gives

$$4a = 4(9.8) - 6v$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$4v\frac{\mathrm{d}v}{\mathrm{d}x} = 39.2 - 6v$$

$$10v\frac{\mathrm{d}v}{\mathrm{d}x} = 98 - 15v$$

$$\left(\frac{10v}{98-15v}\right)\frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

(b) Separating variables in the equation from part (a):

$$\int \frac{10v}{98-15v}\,\mathrm{d}v = \int 1\,\mathrm{d}x$$

To integrate, write

$$\frac{10v}{98-15v} = -\frac{2}{3} + \frac{196}{3(98-15v)}$$

Hence

$$x = -\frac{2v}{3} - \frac{196}{45}\ln|98-15v| + c$$

Initially, $x = 0$ and $v = 0$, so

$$0 = -\frac{196}{45}\ln 98 + c \quad\Longrightarrow\quad c = \frac{196}{45}\ln 98$$

The terminal speed is $\frac{98}{15}\,\mathrm{m\,s^{-1}}$. Since the particle accelerates towards this speed from rest, $98-15v>0$, so the modulus may be removed. Substituting for c and combining the logarithms:

$$\begin{aligned} 45x &= -30v - 196\ln(98-15v) + 196\ln 98 \\ &= 196\ln\left(\frac{98}{98-15v}\right) - 30v \end{aligned}$$

(c) The acceleration is downwards throughout the motion. From Newton's second law,

$$a = 9.8 - 1.5v$$

When its magnitude is $2.45\,\mathrm{m\,s^{-2}}$,

$$2.45 = 9.8 - 1.5v \quad\Longrightarrow\quad v = 4.9\,\mathrm{m\,s^{-1}}$$

Using part (b), the distance descended is

$$\begin{aligned} 45x &= 196\ln\left(\frac{98}{98-15(4.9)}\right) - 30(4.9) \\ x &= \frac{196\ln 4 - 147}{45}\,\mathrm{m} \end{aligned}$$

Let W be the positive work done against resistance. The work done by resistance is $-W$. The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Taking gravitational potential energy to be zero at the final position:

$$\begin{aligned} 0 + 4gx - W &= \frac{1}{2}(4)v^2 + 0 \\ W &= 39.2x - 2v^2 \\ &= 39.2\left(\frac{196\ln 4 - 147}{45}\right) - 2(4.9)^2 \\ &\approx 60.6\,\mathrm{J} \qquad \text{(3 significant figures)} \end{aligned}$$

10. A particle P, of mass m, moves on a rough straight track inclined at an angle α to the horizontal. A fixed point O lies on the track, and x denotes the distance of P from O, measured up the track.

In addition to its weight, P is acted on by a force directed down the track and having magnitude

$$\frac{kmg}{x+a}$$

where a and k are positive constants with dimensions of length. The coefficient of friction between P and the track is μ.

The particle is projected up the track and first comes to rest at the point where $x = 2a$.

(a) Show that, while P is moving up the track with speed v,

$$\frac{\mathrm{d}}{\mathrm{d}x}(v^2) + \frac{2gk}{x+a} + 2g(\sin\alpha + \mu\cos\alpha) = 0 \qquad \textbf{[4]}$$

(b) Verify that

$$v^2 = 2gk\ln\left(\frac{3a}{x+a}\right) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x) \qquad \textbf{[4]}$$

(c) Find, in terms of a, k and α, the range of values of μ for which P remains at rest when $x = 2a$. **[2]**

Solution

(a) Take up the track as the positive direction, and let the normal reaction be R. Resolving perpendicular to the track gives

$$R = mg\cos\alpha$$

While P moves up the track, friction acts down the track with magnitude μR. Using $v\frac{\mathrm{d}v}{\mathrm{d}x}$ for the acceleration, Newton's second law along the track gives

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = -mg\sin\alpha - \mu R - \frac{kmg}{x+a}$$

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -2g(\sin\alpha + \mu\cos\alpha) - \frac{2gk}{x+a}$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}(v^2) = 2v\frac{\mathrm{d}v}{\mathrm{d}x}$, this becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(v^2) + \frac{2gk}{x+a} + 2g(\sin\alpha + \mu\cos\alpha) = 0$$

(b) Integrating the equation in part (a) with respect to x:

$$v^2 = -2gk\ln|x+a| - 2g(\sin\alpha + \mu\cos\alpha)x + c$$

Since x is a distance and $a > 0$, $x + a > 0$, so the modulus can be dropped. Substituting $v = 0$ when $x = 2a$ gives

$$0 = -2gk\ln(3a) - 4ga(\sin\alpha + \mu\cos\alpha) + c$$

$$c = 2gk\ln(3a) + 4ga(\sin\alpha + \mu\cos\alpha)$$

Hence

$$v^2 = 2gk\big(\ln(3a) - \ln(x+a)\big) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x)$$

$$= 2gk\ln\left(\frac{3a}{x+a}\right) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x)$$

(c) At $x = 2a$, let the frictional force acting up the track be F. For equilibrium, Newton's second law gives

$$F - mg\sin\alpha - \frac{kmg}{3a} = 0$$

The required friction must not exceed its limiting value $\mu R = \mu mg\cos\alpha$, so

$$mg\sin\alpha + \frac{kmg}{3a} \leq \mu mg\cos\alpha$$

$$\mu \geq \tan\alpha + \frac{k}{3a\cos\alpha}$$

11. A small instrument package of mass m kg enters a deep liquid moving vertically downwards at $7\,\mathrm{m\,s^{-1}}$. While the package is moving downwards, it experiences a resistive force of magnitude $\frac{m}{20}v^2$ N, where $v\,\mathrm{m\,s^{-1}}$ is its speed.

At time t s after entering the liquid, the package has descended a distance x m.

(a) Find an expression for v in terms of t. **[5]**

(b) Find an expression for x in terms of t. **[4]**

(c) Find the exact distance descended by the package when its speed first reaches $\frac{21}{2}\,\mathrm{m\,s^{-1}}$. **[4]**

Solution

(a) Take vertically downwards as positive. Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - \frac{m}{20}v^2$$

$$20\frac{\mathrm{d}v}{\mathrm{d}t} = 196 - v^2$$

The terminal speed is $14\,\mathrm{m\,s^{-1}}$. Since the initial speed is $7\,\mathrm{m\,s^{-1}}$, the speed increases towards $14\,\mathrm{m\,s^{-1}}$ and remains below it.

Separating variables and using partial fractions:

$$\int \frac{1}{196 - v^2}\,\mathrm{d}v = \frac{1}{20}\int 1\,\mathrm{d}t$$

$$\frac{1}{28}\int\left(\frac{1}{14 - v} + \frac{1}{14 + v}\right)\mathrm{d}v = \frac{t}{20} + c$$

$$\frac{1}{28}(\ln|14 + v| - \ln|14 - v|) = \frac{t}{20} + c$$

Both $14 + v$ and $14 - v$ are positive, so the modulus signs can be dropped. Using $v = 7$ when $t = 0$:

$$c = \frac{1}{28}\ln\frac{21}{7} = \frac{\ln 3}{28}$$

Hence

$$\ln\frac{14 + v}{14 - v} = \frac{7t}{5} + \ln 3$$

$$\frac{14 + v}{14 - v} = 3e^{7t/5}$$

$$14 + v = 42e^{7t/5} - 3ve^{7t/5}$$

$$v\left(1 + 3e^{7t/5}\right) = 14\left(3e^{7t/5} - 1\right)$$

$$v = \frac{14\left(3e^{7t/5} - 1\right)}{3e^{7t/5} + 1}$$

(b) Using $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, rewrite the result from part (a) to recognise a logarithmic integral:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= -14 + \frac{84e^{7t/5}}{3e^{7t/5}+1} \\ &= -14 + 20\frac{\frac{21}{5}e^{7t/5}}{3e^{7t/5}+1} \\ x &= -14t + 20\ln\left(3e^{7t/5}+1\right) + d\end{aligned}$$

Since $x = 0$ when $t = 0$, $d = -20\ln 4$. Therefore

$$x = 20\ln\left(\frac{3e^{7t/5}+1}{4}\right) - 14t$$

(c) Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, the equation of motion becomes

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{196 - v^2}{20}$$

Separating variables and integrating, with $196 - v^2 > 0$:

$$\begin{aligned}\int 1\,\mathrm{d}x &= \int \frac{20v}{196-v^2}\,\mathrm{d}v \\ x &= -10\ln|196 - v^2| + h\end{aligned}$$

At entry, $x = 0$ and $v = 7$, so $h = 10\ln 147$. Thus, when $v = \frac{21}{2}$,

$$x = 10\ln\left(\frac{147}{196 - \left(\frac{21}{2}\right)^2}\right) = 10\ln\left(\frac{12}{7}\right)\text{ m}$$

Final answers with $g = 10$: (a) $v = 10\sqrt{2}\,\frac{(10\sqrt{2}+7)e^{\sqrt{2}t} - (10\sqrt{2}-7)}{(10\sqrt{2}+7)e^{\sqrt{2}t} + (10\sqrt{2}-7)}$;
(b) $x = 20\ln\left(\frac{(10\sqrt{2}+7)e^{\sqrt{2}t} + 10\sqrt{2} - 7}{20\sqrt{2}}\right) - 10\sqrt{2}\,t$; (c) $10\ln\left(\frac{604}{359}\right)$ m

12. A package of mass m kg is released from rest and falls vertically. When its downward speed is $v\,\mathrm{m\,s^{-1}}$, it experiences a resistive force of magnitude mkv^2 N, where k is a positive constant. Take the downward direction as positive.

(a) Show that, at time t s after release,

$$v = \sqrt{\frac{g}{k}}\left(\frac{e^{2\sqrt{gk}\,t} - 1}{e^{2\sqrt{gk}\,t} + 1}\right) \qquad \textbf{[5]}$$

It is given that $k = 0.0125\,\mathrm{m^{-1}}$.

(b) Find the distance fallen by the package during the first 2 seconds after its release. **[4]**

Solution

(a) Taking vertically downwards as positive, Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - mkv^2$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv^2$$

Let $V = \sqrt{g/k}$ be the terminal speed. The package accelerates from rest towards this speed, so $0 \leq v < V$.

Separating variables and using partial fractions:

$$\int \frac{1}{V^2 - v^2}\,\mathrm{d}v = \int k\,\mathrm{d}t$$

$$\frac{1}{2V}\int\left(\frac{1}{V+v} + \frac{1}{V-v}\right)\mathrm{d}v = \int k\,\mathrm{d}t$$

$$\frac{1}{2V}\ln\left|\frac{V+v}{V-v}\right| = kt + c$$

Since $V + v > 0$ and $V - v > 0$, the modulus can be removed. At release, $t = 0$ and $v = 0$, giving $c = 0$. Hence

$$\ln\left(\frac{V+v}{V-v}\right) = 2kVt$$

$$V + v = e^{2kVt}(V - v)$$

$$v\left(1 + e^{2kVt}\right) = V\left(e^{2kVt} - 1\right)$$

$$v = \sqrt{\frac{g}{k}}\left(\frac{e^{2\sqrt{gk}\,t} - 1}{e^{2\sqrt{gk}\,t} + 1}\right)$$

(b) Let s be the distance fallen after t seconds. Using $g = 9.8$ and $k = 0.0125$ in part (a):

$$\frac{\mathrm{d}s}{\mathrm{d}t} = v = 28\left(\frac{e^{0.7t} - 1}{e^{0.7t} + 1}\right)$$

$$= \frac{56e^{0.7t}}{e^{0.7t} + 1} - 28$$

Integrating by recognising the derivative of the denominator:

$$\begin{aligned} s &= \int \left(\frac{56e^{0.7t}}{e^{0.7t}+1} - 28 \right) \mathrm{d}t \\ &= 80 \ln\left|e^{0.7t}+1\right| - 28t + d \end{aligned}$$

Since $e^{0.7t}+1 > 0$, the modulus can be removed. Using $s = 0$ when $t = 0$ gives $d = -80\ln 2$, so

$$s = 80 \ln\left(\frac{e^{0.7t}+1}{2} \right) - 28t$$

The distance fallen in the first 2 seconds is therefore

$$s(2) = \left(80 \ln\left(\frac{e^{1.4}+1}{2} \right) - 56 \right) \text{m} \approx 18.2\,\text{m}$$

Final answers with $g = 10$: (b) $\left(80 \ln\left(\frac{e^{\sqrt{2}}+1}{2} \right) - 40\sqrt{2} \right) \text{m} \approx 18.5\,\text{m}$

13. A particle P of mass 16 kg moves in a straight line on a smooth horizontal plane.

At time $t = 0$, the particle passes through a fixed point O with velocity $\frac{5\sqrt{3}}{4}\,\mathrm{m\,s^{-1}}$.

From this instant, a horizontal force opposing the motion acts on P. The magnitude of the force is proportional to $\sqrt{25 - 4v^2}$, where $v\,\mathrm{m\,s^{-1}}$ is the velocity of P.

When $v = 2$, the magnitude of the force is 6 N. The force ceases to act when the velocity has decreased to $\frac{5}{4}\,\mathrm{m\,s^{-1}}$.

(a) Show that, while the force is acting, the motion of P is modelled by the differential equation

$$\frac{1}{\sqrt{25 - 4v^2}}\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{8} \qquad \textbf{[3]}$$

(b) Hence find v in terms of t, showing that your expression is valid for

$$0 \leq t \leq \frac{2\pi}{3} \qquad \textbf{[5]}$$

(c) Determine the total distance travelled by P during the interval $0 \leq t \leq 8$. Give your answer in an exact form. **[6]**

Solution

(a) Take the initial direction of motion as positive. Let the magnitude of the opposing force be $F = k\sqrt{25 - 4v^2}$, where $k > 0$.

Using the given force when $v = 2$:

$$6 = k\sqrt{25 - 4(2)^2}$$

$$6 = 3k$$

$$k = 2$$

Using $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, Newton's second law in the positive direction gives

$$16\frac{\mathrm{d}v}{\mathrm{d}t} = -2\sqrt{25 - 4v^2}$$

$$\frac{1}{\sqrt{25 - 4v^2}}\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{8}$$

(b) Separating the variables and factoring out the numerical scale before integrating:

$$\frac{\mathrm{d}v}{\sqrt{25 - 4v^2}} = -\frac{1}{8}\,\mathrm{d}t$$

$$\frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{5}{2}\right)^2 - v^2}}\,\mathrm{d}v = -\frac{1}{8}\int \mathrm{d}t$$

$$\frac{1}{2}\arcsin\left(\frac{2v}{5}\right) = -\frac{t}{8} + c$$

Substituting $t = 0$ and $v = \frac{5\sqrt{3}}{4}$:

$$c = \frac{1}{2}\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

Hence

$$\arcsin\left(\frac{2v}{5}\right) = \frac{\pi}{3} - \frac{t}{4}$$

The force ceases when $v = \frac{5}{4}$, so

$$\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{3} - \frac{t}{4}$$

$$\frac{\pi}{6} = \frac{\pi}{3} - \frac{t}{4}$$

$$t = \frac{2\pi}{3}$$

Taking sine gives the required velocity while the force acts:

$$v = \frac{5}{2}\sin\left(\frac{\pi}{3} - \frac{t}{4}\right), \qquad 0 \le t \le \frac{2\pi}{3}$$

(c) Let x be the displacement from O in the positive direction. Using $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ and integrating while the force acts:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{5}{2}\sin\left(\frac{\pi}{3} - \frac{t}{4}\right)$$

$$x = 10\cos\left(\frac{\pi}{3} - \frac{t}{4}\right) + d$$

Since $x = 0$ at $t = 0$,

$$0 = 10\cos\frac{\pi}{3} + d \quad \Longrightarrow \quad d = -5$$

The velocity remains positive, so the distance travelled while the force acts is

$$x\left(\frac{2\pi}{3}\right) = 10\cos\frac{\pi}{6} - 5 = 5\sqrt{3} - 5$$

After the force ceases, there is no horizontal force, so the particle continues at the constant velocity $\frac{5}{4}\,\mathrm{m\,s^{-1}}$. Therefore,

$$\text{Total distance} = (5\sqrt{3} - 5) + \frac{5}{4}\left(8 - \frac{2\pi}{3}\right)$$

$$= \left(5\sqrt{3} + 5 - \frac{5\pi}{6}\right)\mathrm{m}$$

14. A vehicle of mass m kg moves along a horizontal straight road. Its engine works at a constant rate of $2kmg$ W, where k is a positive constant with units of speed. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, it experiences a resistance of magnitude $\dfrac{mgv}{k}$ N.

Initially, the vehicle is at rest at $x = 0$, where x m is the distance travelled.

(a) Show that

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{k}\left(2k^2 - v^2\right) \qquad \textbf{[3]}$$

(b) Verify that

$$gx = -kv + \frac{k^2}{\sqrt{2}}\ln\left(\frac{\sqrt{2}k + v}{\sqrt{2}k - v}\right) \qquad \textbf{[5]}$$

(c) By using the work–energy principle, show that the time taken for the vehicle to accelerate from rest to a speed V, where $0 < V < \sqrt{2}k$, is

$$\frac{k}{2g}\ln\left(\frac{2k^2}{2k^2 - V^2}\right) \qquad \textbf{[8]}$$

Solution

(a) Take the direction of motion as positive. The driving force is power divided by speed:

$$F = \frac{P}{v} = \frac{2kmg}{v}$$

Newton's second law in this direction gives

$$ma = \frac{2kmg}{v} - \frac{mgv}{k}$$

$$a = \frac{g}{kv}\left(2k^2 - v^2\right)$$

Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{kv}\left(2k^2 - v^2\right)$$

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{k}\left(2k^2 - v^2\right)$$

(b) Separating the variables in part (a):

$$g\,\mathrm{d}x = k\frac{v^2}{2k^2 - v^2}\,\mathrm{d}v$$

Use algebraic division followed by partial fractions:

$$\frac{v^2}{2k^2 - v^2} = -1 + \frac{2k^2}{2k^2 - v^2}$$

$$= -1 + \frac{k}{\sqrt{2}}\left(\frac{1}{\sqrt{2}k - v} + \frac{1}{\sqrt{2}k + v}\right)$$

Integrating gives

$$gx = k\int\left[-1+\frac{k}{\sqrt{2}}\left(\frac{1}{\sqrt{2}k-v}+\frac{1}{\sqrt{2}k+v}\right)\right]\mathrm{d}v$$
$$= -kv+\frac{k^2}{\sqrt{2}}\Big(\ln|\sqrt{2}k+v|-\ln|\sqrt{2}k-v|\Big)+c$$

The resultant force is zero at the terminal speed $\sqrt{2}k$. During acceleration from rest, $0 \le v < \sqrt{2}k$, so both logarithm arguments are positive and the moduli can be removed.

When $x = 0$, $v = 0$, giving

$$0 = \frac{k^2}{\sqrt{2}}\Big(\ln(\sqrt{2}k)-\ln(\sqrt{2}k)\Big)+c \quad\Longrightarrow\quad c = 0$$

Combining the logarithms:

$$gx = -kv+\frac{k^2}{\sqrt{2}}\ln\left(\frac{\sqrt{2}k+v}{\sqrt{2}k-v}\right)$$

(c) Let $W(v)$ be the work done by resistance from rest until the speed is v. This work is negative because resistance opposes the motion. From part (a),

$$\mathrm{d}x = \frac{kv^2}{g(2k^2-v^2)}\,\mathrm{d}v$$

Hence

$$\mathrm{d}W = -\frac{mgv}{k}\,\mathrm{d}x$$
$$\frac{\mathrm{d}W}{\mathrm{d}v} = -\frac{mv^3}{2k^2-v^2} = mv-\frac{2mk^2v}{2k^2-v^2}$$

Integrating by recognising the derivative of $2k^2 - v^2$:

$$W(v) = \frac{1}{2}mv^2+mk^2\ln|2k^2-v^2|+d$$

Using $W(0) = 0$ gives $d = -mk^2\ln(2k^2)$. Since $0 < V < \sqrt{2}k$, we have $2k^2 - V^2 > 0$, so

$$W(V) = \frac{1}{2}mV^2-mk^2\ln\left(\frac{2k^2}{2k^2-V^2}\right)$$

The work–energy principle, with work positive when a force acts in the direction of motion, is

$$KE_0+GPE_0+\sum W_{\text{other}} = KE_1+GPE_1$$

Let the required time be t. The engine does work $Pt = 2kmg\,t$, and the gravitational potential energy is unchanged. Taking it to be zero:

$$0+0+2kmg\,t+W(V) = \frac{1}{2}mV^2+0$$
$$2kmg\,t+\frac{1}{2}mV^2-mk^2\ln\left(\frac{2k^2}{2k^2-V^2}\right) = \frac{1}{2}mV^2$$
$$2kmg\,t = mk^2\ln\left(\frac{2k^2}{2k^2-V^2}\right)$$
$$t = \frac{k}{2g}\ln\left(\frac{2k^2}{2k^2-V^2}\right)$$